

Emergent

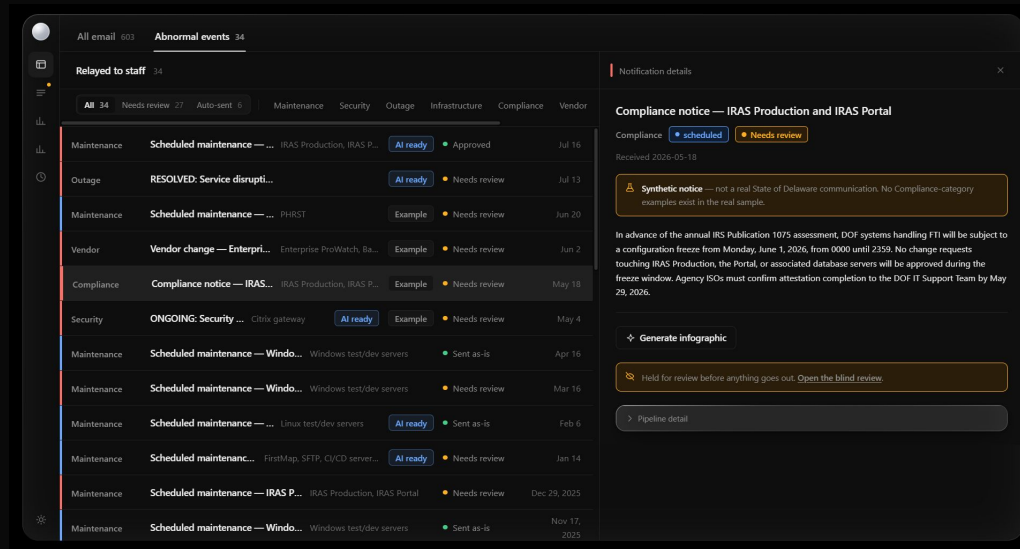
Email delivery **reinvented.**



Too many critical and abnormal notifications. Too little attention.

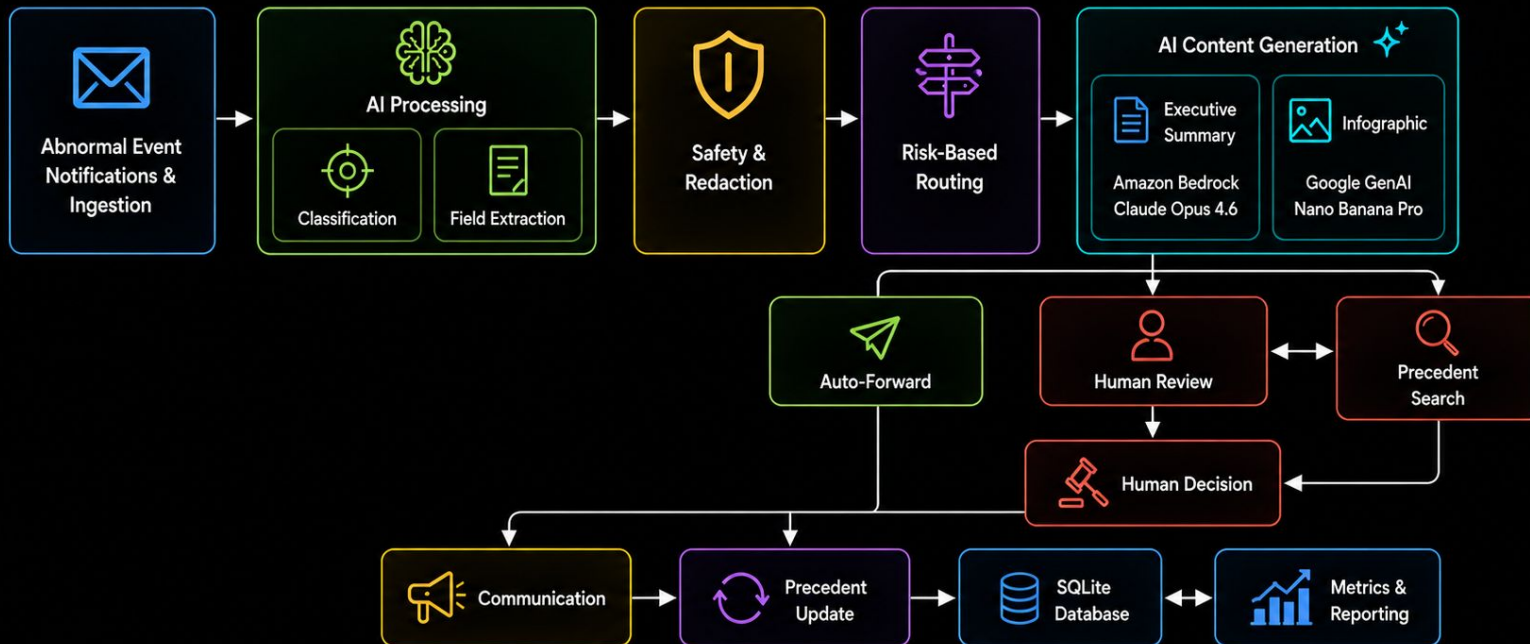
- **600 notifications in just 2 weeks**
Employees get buried in abnormal-event emails on top of their normal workload.
- **Notification fatigue**
When alerts never stop, they blur together. Important ones get skimmed, ignored, or missed.
- **Small interruptions add up**
Even **1–2 minutes** per notification means roughly **10–20 hours** spent reading, interpreting, and figuring out what matters across 600 messages—instead of doing core work.
- **Missed alerts have consequences**
A missed notice means skipped required actions, systems left running through maintenance, or being caught off guard by an outage.

An AI pipeline that classifies incoming IT notices, drafts employee and leadership-facing comms, and routes everything through genuine human approval.



It matches how a reviewer already thinks: rules for what's certain, judgment for what isn't.

Emergent – AI-Powered Abnormal Event Notification System



Next.js



React



TypeScript



Tailwind CSS



SQLite



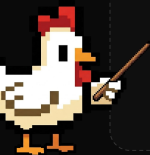
Amazon Bedrock
Claude Opus 4.6



Google GenAI
Nano Banana Pro

Live

Live Demo: <https://emergent0.vercel.app/>



→ Category + confidence → Auto-send or hold

Responsible AI & Data

DATA USED

- 603 real subject lines (1 week, tagged)
- 226 full real notices (19 months)

KNOWN LIMITATIONS

- Every accuracy number here is model-generated labels agreeing with model-generated answers
- Not all real data has been blind-labeled by humans
- A true accuracy figure compared to human labels comes with usage
- Tradeoff: Higher-quality visual generation can require more capable models, more iteration, and greater compute/cost, making fully generated text-heavy graphics challenging

SAFEGUARDS BUILT BEYOND AN API CALL

- For High risk messages, AI-generated content is never auto-sent
- No prompt injection→A trained classifier with real weights, not a prompted LLM
- Enforced in the routing logic and again at the approval endpoint. not as a setting anyone can turn off

SYNTHETIC COVERAGE

[illegible]

 Abnormal Events_004.xlsx Excel Spreadsheet	
Date	Notes
01-01-20	AMS production and the Partial will be undergoing maintenance from Sunday, January 19th, starting from 6:00 AM to 10:00 AM. All orders should be held until the completion of the maintenance.
01-02-20	On Sunday, January 26, 2020, from 10:00 AM to 11:00 AM, there will be performance maintenance on the AMS production Line series which will impact the available production applications.
01-03-20	On Monday, January 31, 2020, from 10:00 AM to 11:00 AM, there will be performance maintenance on available series. To avoid a partial event, AMS will be closed for the day.
01-04-20	On Monday, January 31, 2020, from 10:00 AM to 11:00 AM, there will be performance maintenance on the AMS production Line series which will impact the available production applications. AMS will be closed for the day.
01-05-20	AMS production and the Partial will be undergoing maintenance from Monday, January 31st, starting from 6:00 AM to 10:00 AM. All orders should be held until the completion of the maintenance.
01-06-20	AMS production and the Partial will be undergoing maintenance from Monday, January 31st, starting from 6:00 AM to 10:00 AM. All orders should be held until the completion of the maintenance.
01-07-20	AMS production and the Partial will be undergoing maintenance from Monday, January 31st, starting from 6:00 AM to 10:00 AM. All orders should be held until the completion of the maintenance.
01-08-20	AMS production and the Partial will be undergoing maintenance from Monday, January 31st, starting from 6:00 AM to 10:00 AM. All orders should be held until the completion of the maintenance.
01-09-20	AMS production and the Partial will be undergoing maintenance from Monday, January 31st, starting from 6:00 AM to 10:00 AM. All orders should be held until the completion of the maintenance.
01-10-20	AMS production and the Partial will be undergoing maintenance from Monday, January 31st, starting from 6:00 AM to 10:00 AM. All orders should be held until the completion of the maintenance.
01-11-20	AMS production and the Partial will be undergoing maintenance from Monday, January 31st, starting from 6:00 AM to 10:00 AM. All orders should be held until the completion of the maintenance.
01-12-20	AMS production and the Partial will be undergoing maintenance from Monday, January 31st, starting from 6:00 AM to 10:00 AM. All orders should be held until the completion of the maintenance.

Raw data is messy and unevenly biased. To level unrepresentative bias and obtain enough data for accurate fine-tuning, we created synthetic data for validation and training.

FOR PRODUCTION

- A human-validated labeling pass
- Scoped mailbox integration
- A named approval owner
- Monitoring for reviewer disagreement over time



Compute and Providers



SERVICES USED

AWS Bedrock, Kiro, Vercel, Claude, Github Repository, and Nano Banana through Google Cloud

WHAT RAN THERE

API integration, Application Hosting & Demo Support, Code Refinement, Infographic Design, and GitHub Code Management

WORKED / DIDN'T

- Kiro ↔ Google Cloud setup was difficult: repeated authorization, command, and VM connection failures
- Image-generation limitations: Bedrock image outputs sometimes produced blurry text, misspellings, altered wording, and inconsistent layout

WHAT WE LEARNED

A lot of services have already been thought out and engineered, so instead of reinventing the wheel, it's better to build off of the design and work with all the wonderful providers we have the honor to be with.

WHY EMERGENT?

- Greater awareness of critical events
- Faster understanding, fewer missed details
- Human accountability stays in the loop
- Less disruption to employees' core work

WHO BENEFITS?

- Employees and IT / DTI teams

TIME • COST • QUALITY

- Less drafting, summarizing & formatting
- Staff effort shifts from repetitive communication work to higher-value tasks
- More consistent messages, clearer actions & fewer missed details

PATH TO PILOT

- Start with one agency, Use approved templates + historical notifications, Assign a human reviewer / approver, Measure time saved, accuracy & communication quality

- **PROBLEM**

- Too much information, too little attention

- **SOLUTION**

- AI turns noise into what matters
Classify, extract, summarize, personalize, and keep humans in control.

- **RESULT**

- Clearer communication, faster understanding
Important information becomes easier to find, understand, and act on.

- **IMPACT**

- Less time managing information. More time doing meaningful work.
For Delaware, that means more productive employees and fewer missed critical alerts.

- **ONE STEP FURTHER**

- We started by helping Delaware employees miss fewer critical alerts. The bigger opportunity is helping everyone spend less time managing information, and more time on what matters.
-

emergent.

Any questions? Thank you for your time.

AAYUSH, AIDEN, AUSTIN, MOHADESEH, AND SNEHAL



Content rules that allow an email to pass through

01 Touches a production system, not test or dev

02 Unplanned, an active incident, not scheduled work

03 Security-related in any way

04 Missing a clear start or end date

CLEARs ALL FOUR

Eligible for auto-forward, and only the original untouched email, never anything AI-drafted.

EVIDENCE

Moving the confidence gate 0.55 → 0.70 → 0.80 barely changed outcomes, demonstrating rules were already doing the filtering.

03 – HOW THE CLASSIFIER WAS TRAINED

GENERATE

458 synthetic notices

Balanced across 8 categories, written by 5 model families so no one model's blind spot becomes the dataset's.



LABEL

3-model voting panel

Unanimous votes kept as trustworthy. Split votes treated as ambiguous.



AUDIT & REPAIR

Label Drift: 50% → 78.8% match

Length normalization: length-only model hit 21.9% vs 12.5% chance. Fixed with three forced length bands.

EMBED

Titan → 1,024 numbers

Each email becomes one vector.



TRAIN & SELECT

6 algorithms, cross-validated

Logistic regression won. Selected on training folds only.



8 INDEPENDENT SCORERS

Outage	0.91
Maintenance	0.34
Security	0.12
Infrastructure	0.09
+ 4 more	

Each scorer is one row of 1,024 weights answering one question: how much does this look like my category?

EVALUATE ONCE

Model only sees the real goldset of 226 emails via Jay's inbox one time, stopping possible data leakage.

226 real emails since beginning of 2025

97.8%

Classification agreement, against an 88.9% baseline

0.86

Outage recall — 19 of 22 caught, vs 0.62 for keyword-only rules

0

Outages, security notices or misclassifications auto-sent

THE FAILURE THIS PREVENTS

Keyword-only rules mislabel 5 of the 7 outages they miss as routine Maintenance — exactly the mistake this project exists to stop.

LIVE DEMO SPLIT

6 auto-sent, 28 held for review.